

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 2

BIOLOGY

Paper 1 Multiple Choice

9744/01

17 September 2021

1 hour

Additional Materials: OMR Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C and D**.

Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate **OMR Answer Sheet**.

Read the instructions on the OMR Answer Sheet very carefully.

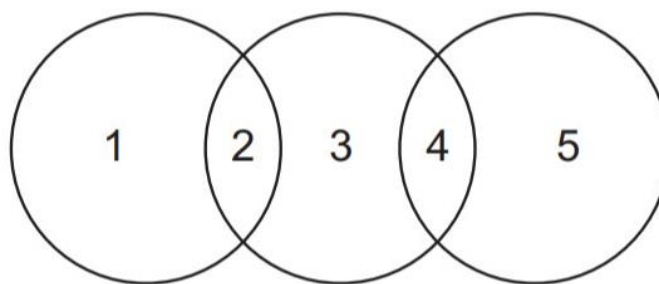
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Answer **all** questions

- 1 The diagram shows some similarities between chloroplasts, mitochondria and typical prokaryotes.



Which row is correct?

	1	2	3	4	5
A	chloroplasts	circular DNA	mitochondria	80S ribosomes	prokaryotes
B	chloroplasts	80S ribosomes	mitochondria	circular DNA	prokaryotes
C	prokaryotes	circular DNA	mitochondria	circular DNA	chloroplasts
D	prokaryotes	70S ribosomes	chloroplasts	80S ribosomes	mitochondria

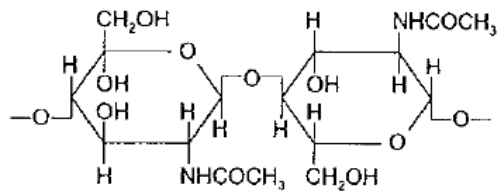
- 2 The table lists the features of two biological molecules found in eukaryotic cells.

Molecule	Exists as a polymer	Hydrogen bonds are formed within the molecule	Remains in the cellular compartment where it is synthesised
X	Yes	Yes	Yes
Y	No	No	No

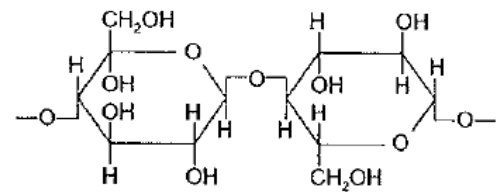
Which of the following shows the possible identities of molecules X and Y?

	Molecule X	Molecule Y
A	Triglyceride	Cellulose
B	Deoxyribonucleic acid	Collagen
C	Haemoglobin	Phospholipid
D	RNA polymerase	Cholesterol

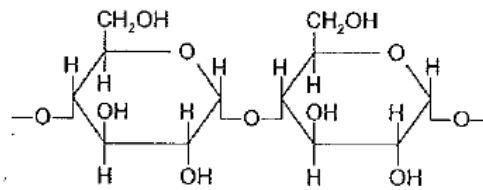
3 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



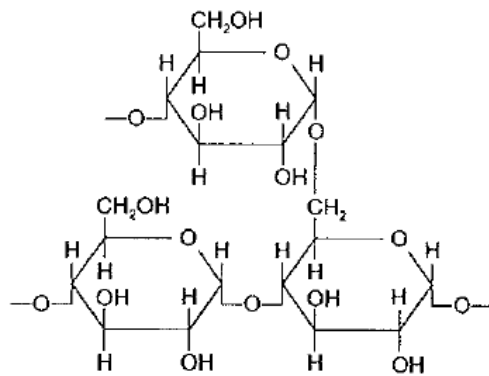
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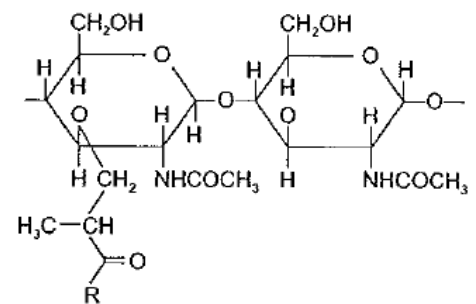
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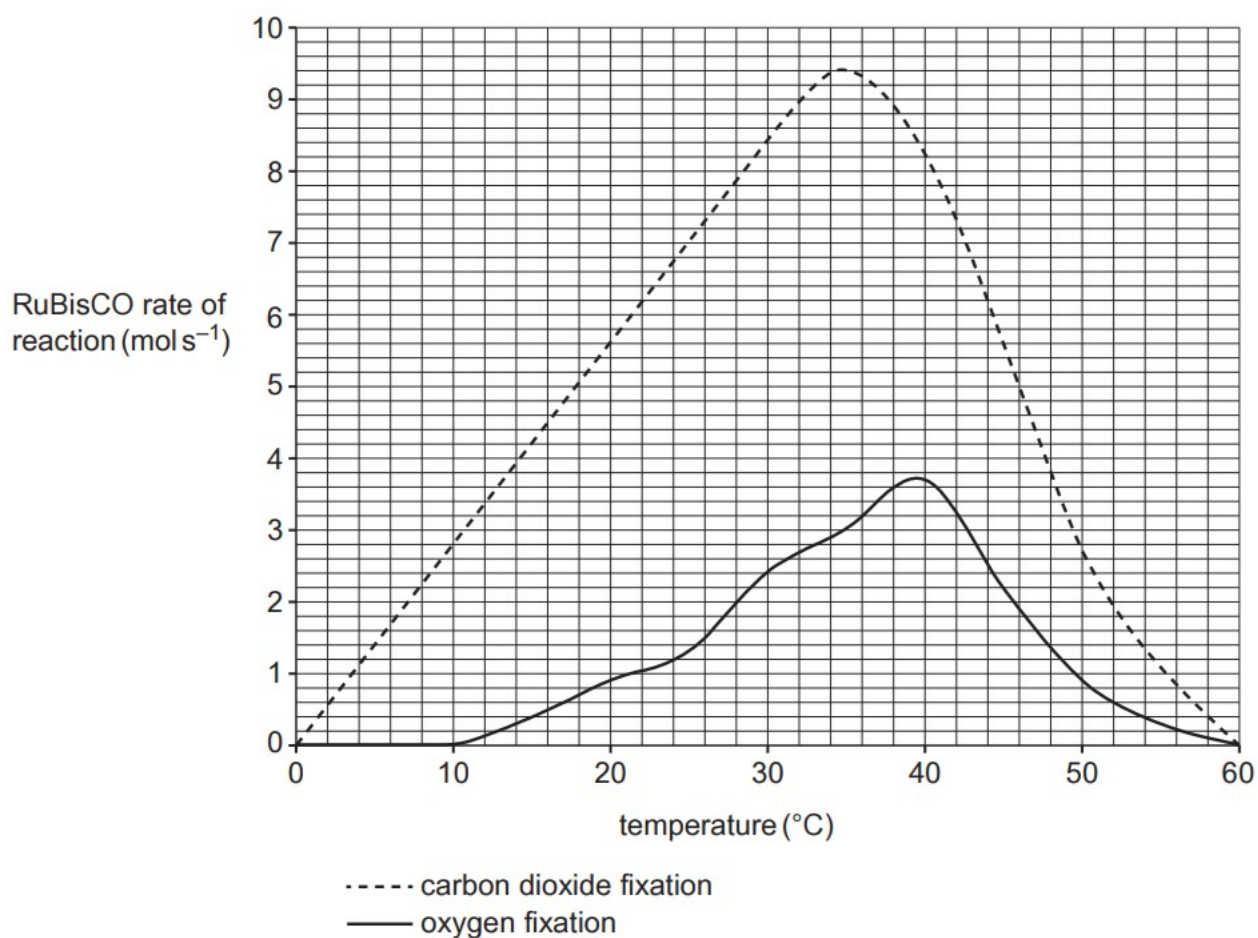
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The polysaccharides can be described as below:

Which shows the correct pairings of the polysaccharide names and diagrams?

	Cellulose	Amylopectin	Peptidoglycan	Amylose	Chitin
A	1	5	4	2	3
B	2	4	1	3	5
C	3	5	2	4	1
D	2	4	5	3	1

- 4 RuBisCO is an enzyme that fixes carbon dioxide in photosynthesis. In some conditions, RuBisCO also carries out oxygen fixation. The graph below shows how the carbon dioxide and oxygen fixing activities of RuBisCO are affected by temperature



What are the correct percentage changes in RuBisCO carbon dioxide and oxygen fixing activities between 30°C and 40°C?

- A carbon dioxide fixation 12.7%, oxygen fixation 23.3%
- B carbon dioxide fixation 14.6%, oxygen fixation 18.9%
- C carbon dioxide fixation 2.4%, oxygen fixation 54.2%
- D carbon dioxide fixation 3.6%, oxygen fixation 35.1%

- 5** Batrachotoxin is a poison found in frogs in the Columbian jungle. The poison is used by Native Indians to produce poison darts. The poison works by increasing the permeability of some cell surface membranes to sodium ions, which move out of the cells.

Which statements are correct for cells affected by batrachotoxin?

- 1 The intracellular fluid has a less negative water potential than the extracellular fluid.
- 2 The extracellular fluid has a less negative water potential than the intracellular fluid.
- 3 Water leaves the cells by osmosis, causing the cells to shrink.
- 4 Water enters the cells by osmosis, causing the cells to swell.

- A** 1 and 3
B 1 and 4
C 2 and 3
D 2 and 4

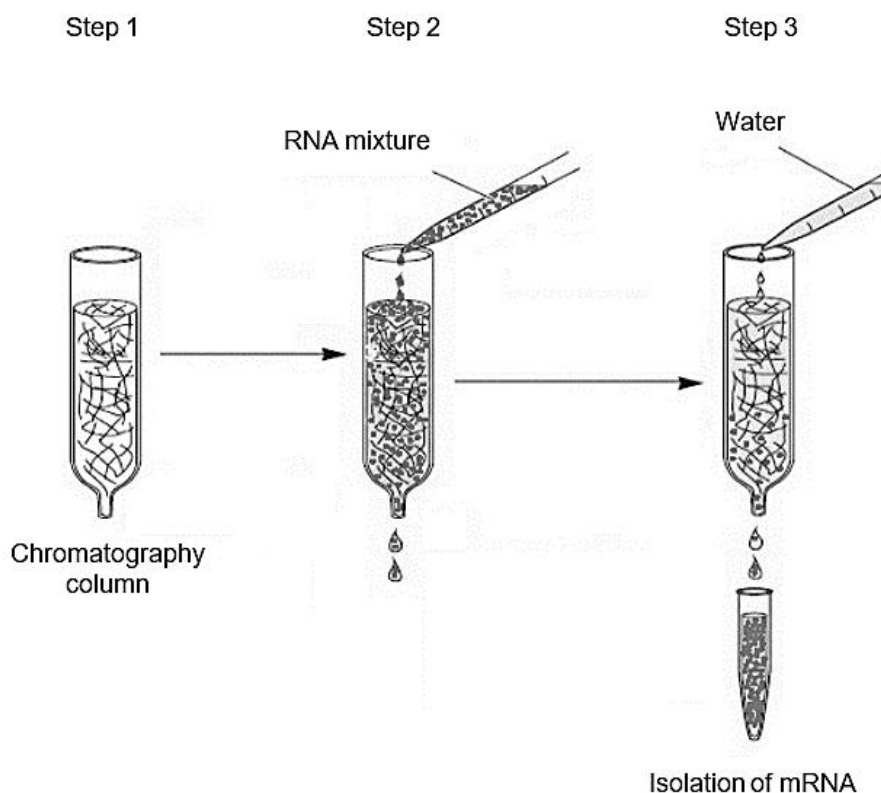
- 6** Induced pluripotent stem cells are stem cells that can be generated directly from differentiated somatic cells under the influence of molecular signals.

Which of the following statements is/are true?

- 1 An induced pluripotent stem cell can become any cell of the developed organism, but cannot produce trophoblast and placenta to support organismal development, whereas a totipotent stem cell can produce a whole organism including extraembryonic tissue.
- 2 A totipotent stem cell and induced pluripotent stem cell can give rise to any cell type, including the extraembryonic membranes.
- 3 An induced pluripotent stem cell can give rise to a single cell lineage whereas a totipotent stem cell can give rise to multiple, but limited number of cell lineages.
- 4 A totipotent stem cell can become any cell of a developed organism, but cannot produce extraembryonic tissue to support organismal development, whereas an induced pluripotent stem cell can produce a whole organism including extraembryonic tissue.
- 5 Induced pluripotent stem cells have the same developmental potential as embryonic stem cells.

- A** 1 only
B 1 and 5 only
C 2, 3 and 4 only
D 3, 4 and 5 only

- 7 The isolation of mRNA can be done by passing an RNA mixture obtained from homogenised tissue through a chromatography column. The procedure of the mRNA isolation is as follows:



- Step 1: Set up a chromatography column which contains short lengths of uracil nucleotides attached to a solid support medium.
- Step 2: Add RNA mixture to the chromatography column. RNA that do not hybridise with the uracil nucleotides will pass through and leave the column.
- Step 3: Add water to the chromatography column to remove and isolate the hybridised mRNA.

Which of the following statements is **not** true?

- A rRNA and tRNA molecules could pass through and leave the column in Step 2
- B Complementary base pairing is responsible for the attachment of RNA to the solid support medium.
- C The isolated mRNA in Step 3 do not contain introns.
- D The RNA that attached to the chromatography column lack poly-(A) tails.

- 8 A polypeptide has the amino acid sequence glycine – arginine – lysine – serine.

The table below gives possible tRNA anticodons for each amino acid.

AMINO ACID	tRNA anticodons	
Arginine	3'UCC5'	3'GCG5'
Glycine	3'CCA5'	3'CCU5'
Lysine	3'UUC5'	3'UUU5'
Serine	3'AGG5'	3'UCG5'

Which sequence of bases on DNA would code for the polypeptide?

- A** 3' CCA CGC AAG AGC 5'
- B** 3' CCT TCC TTC TCG 5'
- C** 5' GGA AGG AAA AGC 3'
- D** 5' GGT TGG TTG TGC 3'
- 9 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

- 10** Which of the following is true of gene regulation in both prokaryotes and eukaryotes?
- A** Gene regulation of eukaryotes involves bending of DNA while in prokaryotes, chemical modifications to DNA is required.
 - B** In eukaryotes, increase in efficiency of RNA polymerase requires specific transcription factors to activate the initiation complex while in prokaryotes, activators are required to bind directly to promoter site to increase efficiency of RNA polymerase.
 - C** In eukaryotes, RNA polymerase binds to the TATA box while in prokaryotes, RNA polymerase binds to the regulatory gene.
 - D** Repressors and activators regulate expression in eukaryotes while prokaryotes are regulated by repressors only.
- 11** Fabry's Disease is a disease that results from a mutation that occurs in the *α -galactosidase A* gene. The sequence of part of the normal and mutated alleles for *α -galactosidase A* gene is shown below.

Normal allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CAG	AGG	UUC	UAA	GGC	GGA

Mutated allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CCG	CAG	AGG	UUC	UAA	GGC

Using the information of the normal and mutated alleles above, it is reasonable to conclude that

- A** A frame shift mutation has occurred.
 - B** A duplication has occurred.
 - C** An insertion of an amino acid has occurred in the mRNA.
 - D** The polypeptide that is translated from the mutated allele will be longer.
- 12** Which of the following statements about the cell cycle is correct?
- A** G1 is relatively the longest phase; proteins, organelles and ATP are synthesised during this phase.
 - B** G2 phase occurs before Interphase of the cell cycle.
 - C** M phase represents the phase where only mitosis occurs.
 - D** G0 occurs between any phases within the cell cycle.

13 Which statement about cancer is **not** correct?

- A** Both alleles for a tumour suppressor gene must undergo loss-of-function mutations in the development of cancer.
- B** Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- C** The growth of new blood vessels is a fundamental step in the development of a malignant tumour.
- D** The risk of cancer is higher when *p53* gene is placed under the control of a strong viral promoter

14 The table below shows the possible combinations of error that lead to cancer and the mode of its inheritance.

Which statement is correct?

	Dysregulation of Cell Cycle	Cause for cancer	Inherited	Mode of inheritance
A	G1 and G2 check point only	Proto-oncogene to oncogene in gametic cells	Yes	Dominant
B	G1 and S check point only	Mutation in <i>p53</i> gene in somatic cells	No	Recessive
C	G1 and G2 check point	Translocation of proto-onco gene to highly expressed promoter in somatic cells	Yes	Dominant
D	G1 and M check point	Non-disjunction of chromosome holding proto-oncogene in gametic cells	No	Recessive

- 15** In the fruit fly, *Drosophila melanogaster*, the wing length can be normal or vestigial (small) while the eye colours can be red or sepia (reddish-brown).

In an investigation, pure-breeding parent flies were crossed. Flies with normal wings and red eyes were crossed with flies with vestigial wings and sepia eyes.

All the F_1 offspring of this cross had normal wings and red eyes.

Flies from this F_1 generation were crossed and the phenotypes of their offspring (F_2 generation) were counted and recorded in the table.

phenotype	observed numbers (O)	expected numbers (E)	$O - E$	$(O - E)^2$	$(O - E)^2 / E$
normal wings and red eyes	885				
normal wings and sepia eyes	322				
vestigial wings and red eyes	286				
vestigial wings and sepia eyes	107				
total	1600	1600	0		$\chi^2 =$

Assuming normal Mendelian inheritance, what is the value of χ^2 to one decimal place?

- A** 0.6
- B** 2.9
- C** 3.0
- D** 850.4

- 16** In cattle, the gene responsible for normal development of hair and teeth, *ectodysplasin 1* (*ED1*) is located on the X chromosome. Mutations in the *ED1* gene result in a rare genetic disorder, anhidrotic ectodermal dysplasia.

Another character, the presence of horns, is determined by a gene on an autosome. The allele for the absence of horns (H) is dominant and the allele for the presence of horns (h) is recessive.

A horned bull with anhidrotic ectodermal dysplasia was mated on several occasions to the same female.

A large number of offspring consisting of males and females in equal numbers in all combinations of phenotypes are shown in the table.

Offspring phenotypes
No anhidrotic ectodermal dysplasia, horns present
No anhidrotic ectodermal dysplasia, horns absent
Anhidrotic ectodermal dysplasia, horns present
Anhidrotic ectodermal dysplasia, horns absent

If X^E represents an X chromosome carrying the normal *ED1* allele and X^e represents an X chromosome carrying the *ED1* allele for anhidrotic ectodermal dysplasia, what is the genotype of the female parent?

- A** $X^E X^E H H$
- B** $X^E X^E H h$
- C** $X^E X^e H H$
- D** $X^E X^e H h$
- 17** A number of different viruses can cause food poisoning.

One virus that can cause food poisoning is the Norovirus. Noroviruses are RNA viruses.

The following four statements are features of viruses:

- 1 have envelopes
- 2 contain reverse transcriptase
- 3 surrounded by a protein coat
- 4 have a helical structure

Which of the following is correct for all RNA viruses:

- A** 1, 2 and 3
- B** 2 only
- C** 3 only
- D** 1 and 4

- 18** *trp* operon is an example of a repressible operon. Which of the following is true about *trp* operon?
- A** *trp* operon contains in total 4 structural genes; namely *trpD*, *trpC*, *trpB* and *trpA*.
 - B** *trp* repressor protein is synthesised in its inactive state, it will be activated upon binding of tryptophan.
 - C** The structural genes in *trp* operon are not transcribed in the absence of tryptophan
 - D** The transcription of structural genes in *trp* operon synthesises multiple monocistronic mRNA where each mRNA codes for one protein.
- 19** A mutation in the *lacI* gene results in the production of a *lac* repressor protein that is unable to bind to allolactose.

Which row correctly shows the level of expression of the *lac* operon under the conditions shown below?

	Lactose	Glucose	Level of expression of <i>lac</i> operon
A	Absent	Absent	Zero
B	Absent	Present	Basal / low
C	Present	Absent	High
D	Present	Present	Basal / low

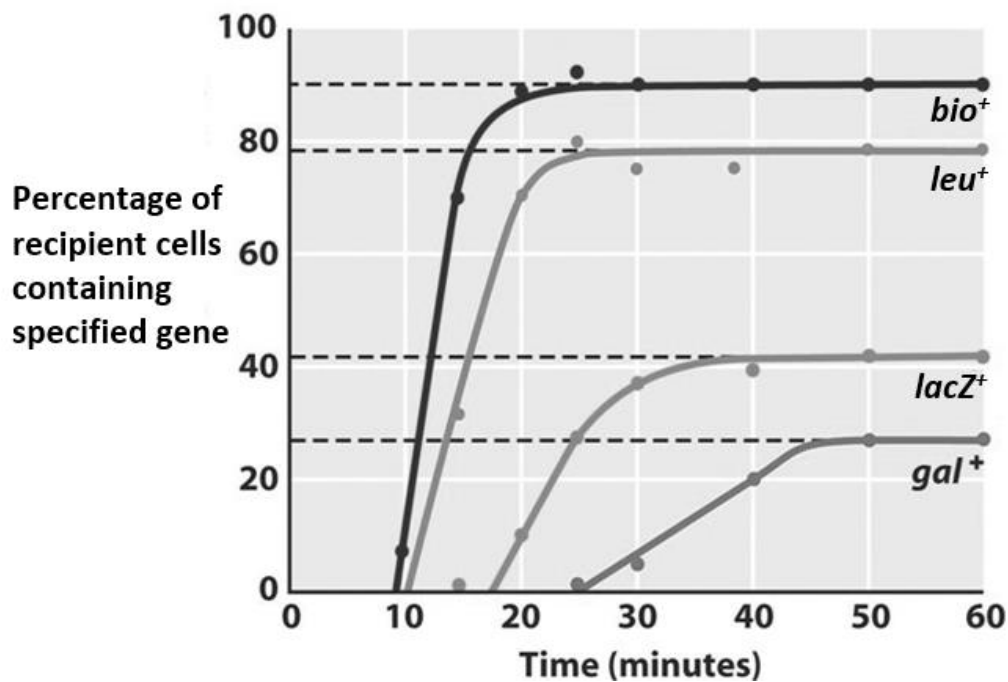
- 20 A plasmid has an origin of transfer (*oriT*) to allow the transfer of DNA from a donor to a recipient cell during bacterial conjugation.

Genes that are closer to *oriT* on the plasmid will be transferred into the recipient cell earlier.

A sample of donor cells was mixed with a sample of recipient cells. The genotypes of these cells are as shown below:

Donor cell	:	<i>bio</i> ⁺	<i>leu</i> ⁺	<i>lacZ</i> ⁺	<i>gal</i> ⁺	Key:	+	: present
Recipient cell	:	<i>bio</i> ⁻	<i>leu</i> ⁻	<i>lacZ</i> ⁻	<i>gal</i> ⁻		-	: absent

Conjugation of bacteria was disrupted at various time points, stopping the transfer of DNA. The recipient cells containing the respective genes after the various time points are shown in the graph.

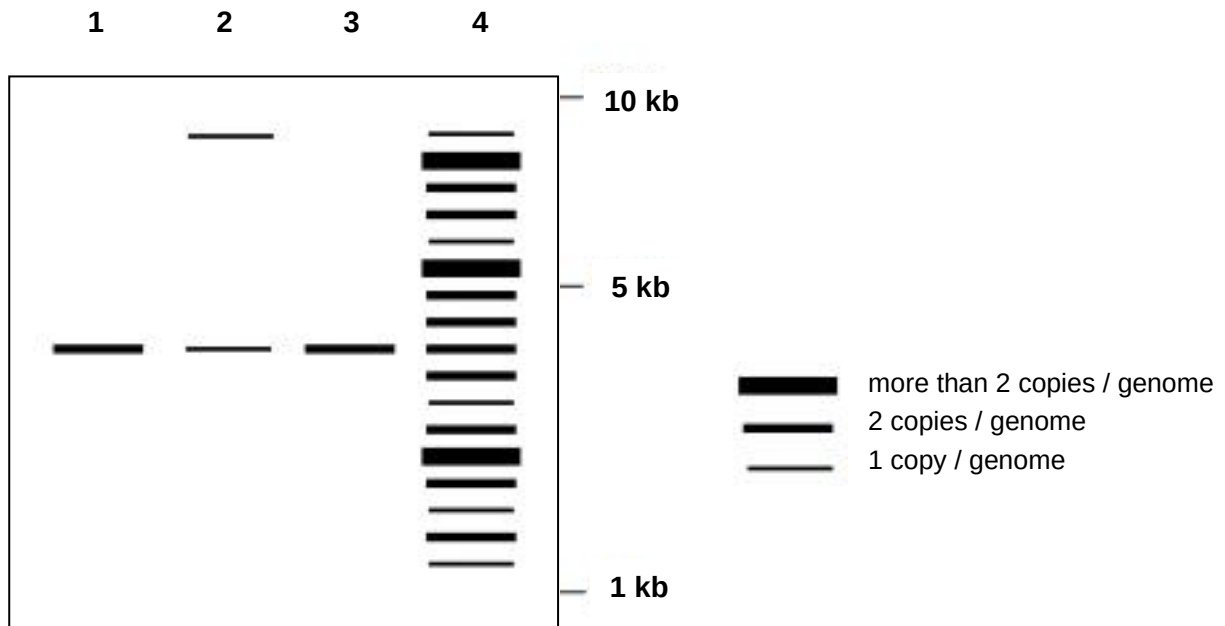


Which statements can be deduced from the information provided?

- 1 Of the four genes, the *gal* gene is the furthest from *oriT*.
 - 2 None of the recipient cells would be able to hydrolyse lactose before 18 minutes if lactose was added to the medium.
 - 3 The *leu* gene is located closer to the *bio* gene than the *lac* gene on the plasmid.
 - 4 The transfer of all four genes to all recipient cells in the sample took 45 minutes.
- A** 1, 2, 3 and 4
B 1, 2 and 3 only
C 1 and 4 only
D 2 and 3 only

- 21** The *ras* oncogene is involved in a variety of human cancers. DNA isolated from a number of normal and cancerous human tissues were digested with the restriction enzyme, *EcoRI*.

A Southern blot was performed and the membrane was probed with radioactive-labelled cloned *ras* DNA. After exposure to x-ray film, the resulting autoradiograph is shown below.



Lane	Sample
1	white blood cells from a healthy human
2	lymphoma cells (cancerous)
3	bladder carcinoma cells (cancerous)
4	sarcoma cells (cancerous)

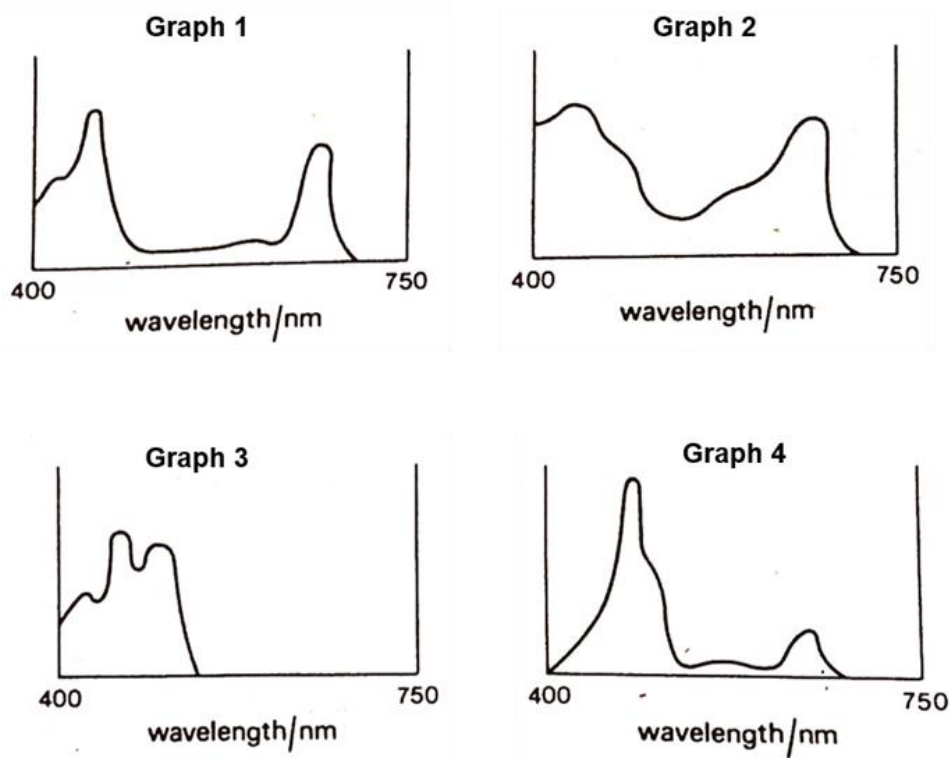
Based on the experimental results, which of the following statement(s) could be deduced?

I	In tissue 3 , there was no mutation within the <i>ras</i> coding region.
II	The result for tissue 2 could be explained by separate point mutations that caused the removal of an <i>EcoRI</i> site and the creation of another within the probe DNA.
III	Gene amplification was involved in tissue 4 .
IV	In tissue 3 , the mutation did not affect existing <i>EcoRI</i> sites within the <i>ras</i> gene.

- A** II only
- B** I, III and IV
- C** II, III and IV
- D** III and IV

- 22** Three of the four graphs below show the absorption spectra of photosynthetic pigments. The other graph shows the action spectrum of photosynthesis for a plant containing those pigments.

The x-axes show wavelength. Three of the y-axes show the percentage light absorbed while the other shows the rate of photosynthesis measured by the rate of evolution of oxygen.



Which one of the following series identifies the four graphs?

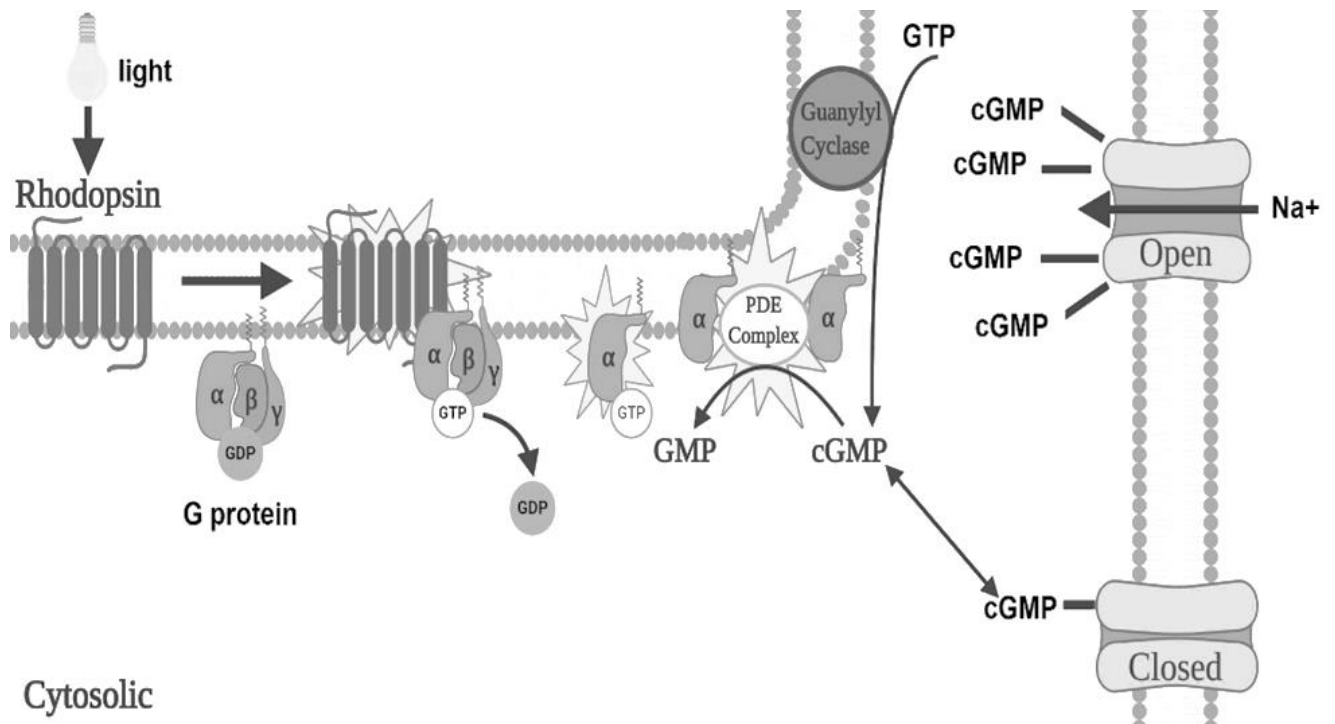
	<i>Absorption spectra</i>			<i>Action spectrum</i>
	Chlorophyll a	Chlorophyll b	Carotenoids	
A	1	4	3	2
B	2	1	3	4
C	4	2	1	3
D	3	2	4	1

- 23** Which of the following statements about aerobic respiration is correct?

- A** NADH and FADH are needed in the processes of respiration within and outside of the mitochondria.
- B** The conversion of succinate to α -ketoglutarate is part of the process that yields NAD during Krebs cycle.
- C** At the end of glycolysis, pyruvate is converted to acetyl CoA in cytoplasm and transported into the mitochondrial matrix.
- D** ATP is generated by substrate level phosphorylation during the formation of succinate.

- 24** Rhodopsin is a G-protein linked receptor in rod bipolar cells in the retina of the eye.

The figure below shows how it is activated, leading to a change in electric charges across the rod cell membrane.



Which statements correctly describe rhodopsin and its cell signalling pathway?

- 1 Rhodopsin is activated by a ligand with complementary shape binding to its ligand binding site.
 - 2 Rhodopsin has amino acid residues that interact with G-proteins.
 - 3 Two G-protein molecules bind to and activate phosphodiesterase (PDE) complex, which leads to decreased cGMP levels inside the cell.
 - 4 Signal reception leads to increased sodium ions inside the cell.
- A** 1 and 2 only
- B** 2 and 3 only
- C** 1, 3 and 4
- D** 1, 2, 3 and 4

25 Researchers have found evidence of natural selection in humans.

- 1 In ancestral populations, only babies and children needed to digest the milk sugar, lactose. The gene coding for the enzyme lactase (*LCT* gene) was switched off before adulthood.
- 2 Today, lactose-intolerant adults cannot digest lactose, leading to abdominal pain after eating food containing lactose.
- 3 A mutation in the *LCT* gene results in adults to be able to digest lactose. This is called lactose persistence.
- 4 The increase in lactose persistence in Europe coincided with an increase in farming of cows for milk.

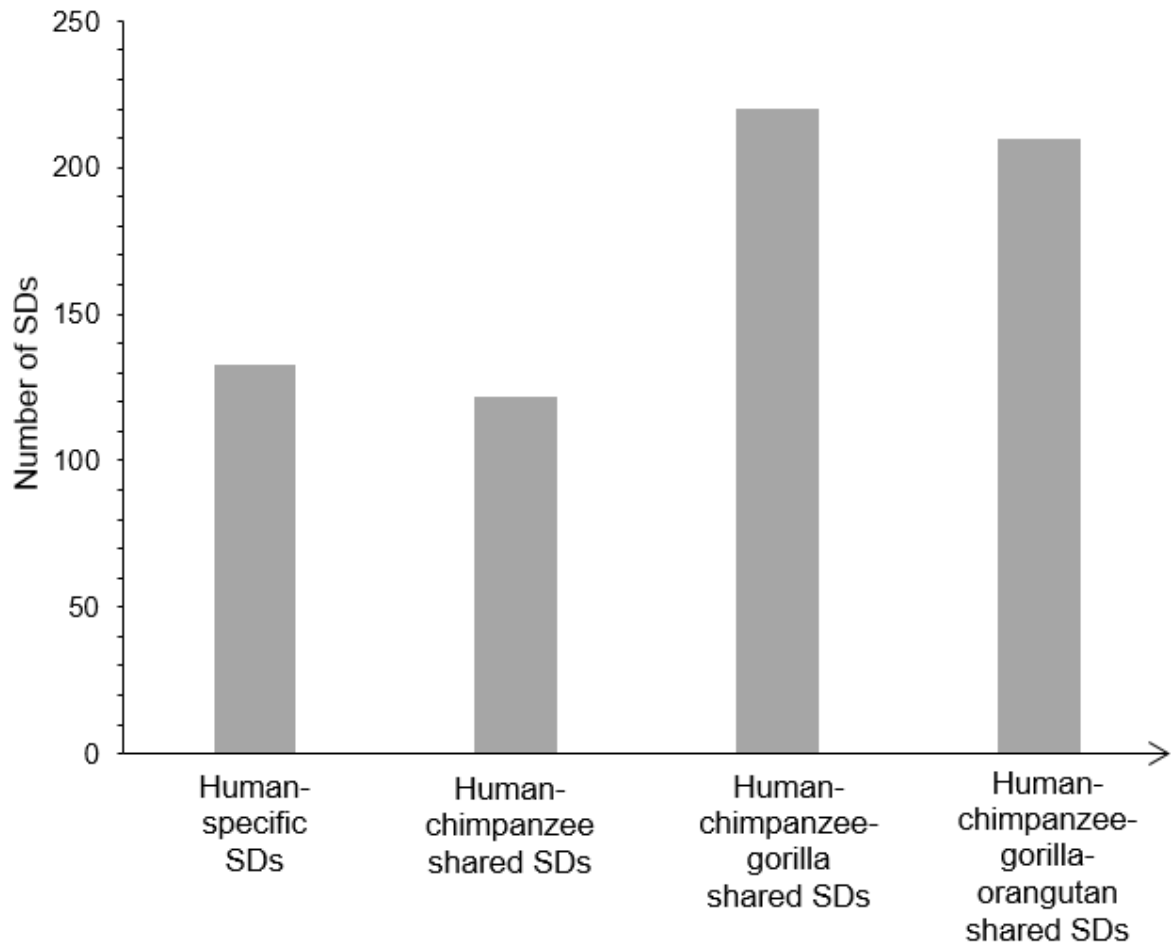
Which statement does **not** explain why there was selection for lactose persistence in humans several thousand years ago?

- A** Reliance on milk products selected for individuals who were able to digest lactose due to the presence of the mutated *LCT* gene in them.
- B** Compared to lactose-intolerant individuals, individuals with lactose persistence are likely to get fewer side effects after consuming food containing lactose.
- C** On average, more individuals who produce high levels of lactase in their lifetime live to reproductive age, as they were able to utilise an additional source of nutrient.
- D** Individuals with lactose persistence consumed more milk than the rest of the population and passed on the mutant allele to their offspring.

- 26** Analysis of the genome of primates reveals many DNA segments that have been duplicated known as segmental duplications (SDs).

It is possible to deduce the phylogenetic relationships of various primates by studying the SDs. Human-specific SDs occurred after humans and chimpanzees diverged. Common SDs shared between humans and chimpanzees (human-chimpanzee shared SDs) occurred after their common ancestor diverged from gorillas and so on.

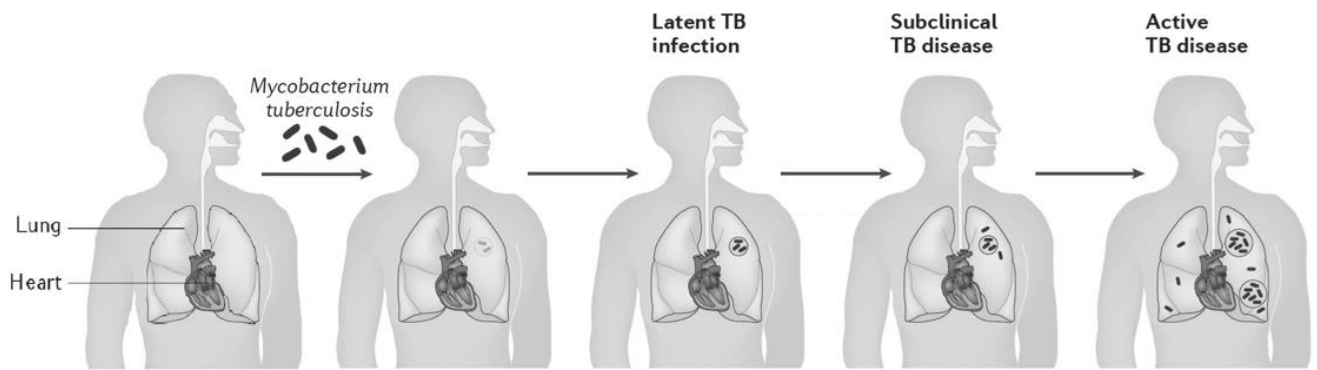
The chart shows the number of SDs in different groups of primates.



Which statement(s) could be deduced from the chart above?

- 1 The number of SDs that can be found in humans but not in other primates is 133.
 - 2 Since the separation of orangutans from other primates, the number of shared SDs present in the other primate group decreases.
 - 3 It is possible for a few SDs to be found in humans and gorillas but not chimpanzees because the same SDs occurred independently in both humans and gorillas.
 - 4 Orangutans are more closely related to chimpanzees than they are to gorillas.
- A** 1, 2 and 4
- B** 2 and 3 only
- C** 3 and 4 only
- D** 1 only

- 27 The diagram shows the exposure of a person to *Mycobacterium tuberculosis* and subsequent progression of tuberculosis (TB).



Which of the statements are **not** correct?

- 1 When an infected person is in the latent TB infection stage, transmission of *M. tuberculosis* to others is unlikely to occur.
- 2 At the centre of the tubercle, helper T cells release *M. tuberculosis* into the cavity.
- 3 Phagocytosis of *M. tuberculosis* by macrophages starts in the subclinical TB disease stage.
- 4 In the active TB disease stage, an infected person's lungs will be destroyed progressively as more tubercles rupture.

- A 1 and 3
 B 1 and 4
 C 2 and 3
 D 2 and 4

- 28 Which row does **not** correctly describe innate and adaptive immunity?

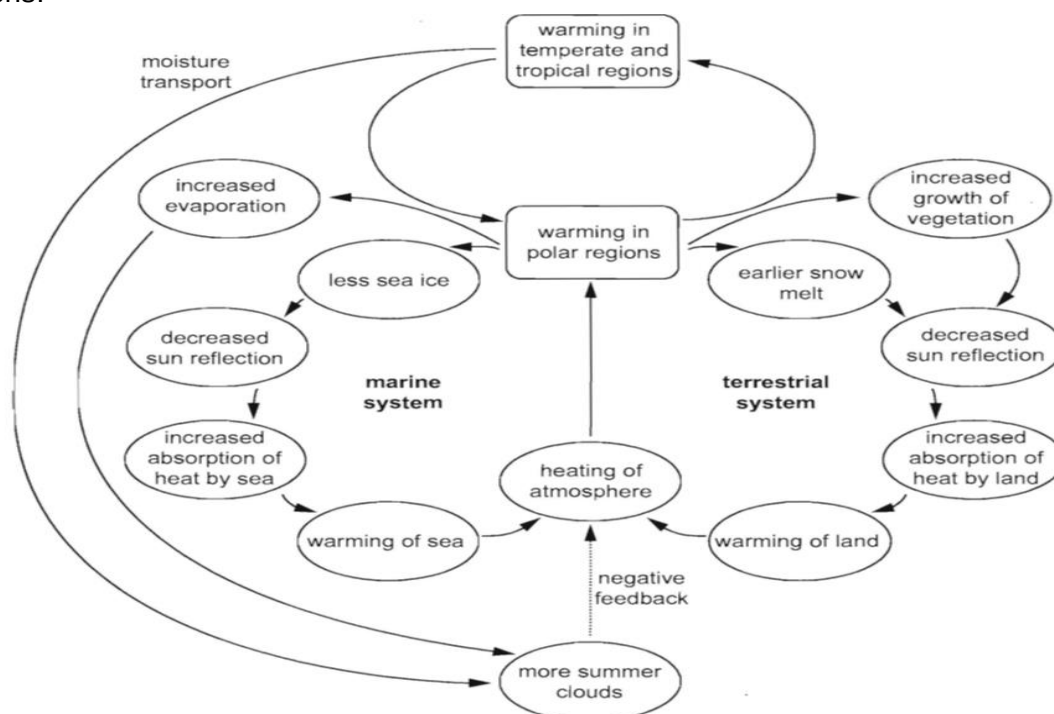
	Innate immunity	Adaptive immunity
A	Comprises humoral immunity and cell-mediated immune response	Involved in inflammatory responses
B	Immediate response	Delayed response
C	Activated by antigens found on many pathogens	Activated by antigens found on specific pathogens
D	Involves mast cells and macrophages	Involves T and B lymphocytes

29 Global warming is having a major impact on species distribution of flora and fauna across the world. Which of these are likely to occur as a result of global warming?

- 1 Increase in number of pests and pathogens in higher latitudes.
- 2 Migration of temperature plants to higher altitudes.
- 3 Decrease in the rate of decomposition.
- 4 Pollinators are more in sync with plant flowering times.
- 5 Keystone species in the tropics become endangered and risk extinction.

- A** 1, 2 and 4 only
- B** 1, 2 and 5 only
- C** 1, 2, 3 and 4 only
- D** 1, 2, 3, 4 and 5

30 The diagram shows the effect of increasing temperatures on the ice and snow cover at the Polar Regions.



Which effect of higher temperatures in the Polar Regions could increase global warming?

- A** Increased evaporation leads to more rainfall, which absorbs heat from the land and sea.
- B** Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.
- C** Melting of sea ice causes more cloud formation, which increases absorption of heat in the atmosphere.
- D** Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.